

Course Outline

COURSE INFORMATION

Faculty/School Name	School of Mathematical and Computational Sciences
Course Name	Single Variable Calculus II
Course Code	Math 1920
Semester	Fall 2025
Academic Year	2025 – 2026

INSTRUCTOR CONTACT INFORMATION

Instructor(s) Name	Waled Emam
E-mail	waled.emam@uofcanada.edu.eg
Office Location	B-311
Office Hours	Monday 15:30 – 16:15 pm

TEACHING ASSISTANT (TA) CONTACT INFORMATION (IF APPLICABLE)

Teaching Assistant Name	Mr. Omar Eltouny
E-mail	Omar.Eltouny@uofcanada.edu.eg
Office Location	
Office Hours	Monday 15:30 – 16:15 pm

COURSE TIME & LOCATION

Class Session	Time	Location
Lecture 1	Monday 13:40-14:55 pm	B105
Lecture 2	Tuesday 13:40-14:55 pm	B105
Lecture 3	Tuesday 15:00-16:15 pm	B105
Lab/tutorial (if applicable)	Tuesday 9:50 - 11:05 am	A412

Course Outline

COURSE DESCRIPTION, CREDITS & PREREQUISITES

Course Description	<i>This course is a continuation of integral calculus of functions of a single variable and an introduction to sequences and series. Techniques of integration are studied, including improper integrals and numerical integration, and the tools of integral calculus are used to compute areas, volumes and arc lengths; and are applied to problems in physics and differential equations. Sequences, series, tests for convergence, Taylor series and Taylor polynomials are studied. Three lecture hours per week.</i>
Course Credits	Three lecture hours per week.
Prerequisite/s	Math 1910 - Must be completed prior to taking this course. Math-1920T - Must be taken at the same time as this course.
Passing grade/prerequisite grade	

REQUIRED TEXTBOOK

COURSE LEARNING OBJECTIVES

1. Apply integration techniques to evaluate definite and indefinite integrals, including improper integrals.
2. Apply integration to evaluate areas and volumes of 2D and 3D regions described by algebraic and transcendental functions, including inverse trigonometric and hyperbolic function.
3. Apply various methods and tests to determine the convergence of sequences and series.
4. Use the tools of infinite series to calculate certain approximations.
5. Prove or disprove statements about integration techniques and applications, as well as infinite sequences

GRADING SCHEME

UPEI Grade Scale can be accessed at: <https://calendar.upei.ca/current/chapter/undergraduate-and-professional-programs-academic-regulations/>

<i>Assessment</i>	<i>Assessment</i>	<i>Weight</i>	<i>Date</i>
1	Quiz 1	5%	14 October 2025
2	Assignment 1	10%	28 October 2025
3	Exam 1	15%	28 October 2025

Course Outline

4	Quiz 2	5%	11 November 2025
5	Assignment 2	10%	25 November 2025
6	Exam 2	15%	25 November 2025
7	Quiz 3	5%	2 December 2025
8	Assignment 3	10%	9 December 2025
9	Final Exam	30%	TBA
Total		100	

CLASS SCHEDULE

<i>Week</i>	<i>Lecture</i>	<i>Lab/ Tutorial (if applicable)</i>
1	Substitution Method	Worksheet 1
2	Integration by Parts	Worksheet 2
3	Trigonometric Integrals	Worksheet 3
4	Trigonometric Substitutions	Worksheet 4
5	Integration by Partial Fractions	Worksheet 5
6	Applications of Definite Integrals (Volume and Length Computations)	Worksheet 6
7	Improper Integrals and Hyperbolic Functions	Worksheet 7
8	Sequences and Sum of a Series	Worksheet 8
9	Test for Divergence and Integral Test	Worksheet 9
10	Comparison Test and Alternating Series Test	Worksheet 10
11	Absolute and Conditional Convergence and The Ratio and Ratio Tests	Worksheet 11
12	Radius of Convergence of Power Series and Taylor and Maclaurin Series	Worksheet 12

Course Outline

UNIVERSITY POLICIES

Attendance and Punctuality

Students are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, tutorials / labs and mandatory office hours for probation & Gateway students. Accordingly, students should note the following:

1. The instructors are not required to substitute assignments or do make-up examinations for absent students.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.
3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for a course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency.

Code of Conduct for Students

The University of Prince Edward Island expects all students to comply with all University regulations and policies, all government laws, and any codes of ethics that govern students who are members of regulated professions. All students are strongly advised to become familiar with this code, which can be accessed at: [UofCanada Student Code of Conduct \(Non-Academic\).pdf](#)

Academic Integrity

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be accessed at: [UPEI Cairo Academic Integrity Policy.pdf](#)

The Use of AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only upon receiving written approval from the instructor. Students should disclose in their assignments the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered academic dishonesty.

Course Outline

Student Accommodation

The university strives to make the learning experience as accessible and inclusive as possible. If you have accessibility needs that require academic accommodation, kindly contact the student wellness department and review UPEI Cairo campus academic accommodation policy which can be accessed at: [UPEI Cairo Campus Academic Accommodations Policy.pdf](#)

Safety Guideline

The University is committed to maintaining a safe and healthy environment for all members while fostering a strong culture of health and safety. Faculty are responsible for promptly and effectively addressing unsafe situations. All faculty, staff, and students must be familiar with and adhere to the safe work procedures management which can be accessed at: [Operational Health and Safety Manual.pdf](#)

Note: Non-compliance with safety regulations may result in a **3-5% deduction** from the course grade per incident, in addition to any other consequences outlined in the safety regulations.

EXPECTATIONS FORM STUDENTS

- 1- There will be no make-up exams allowed without a valid reason or clinic approval.
- 2- All reports required during this course shall be original and adhere to the university's academic integrity policy. Reports will be submitted through Turnitin, which checks for similarities with other submissions and online sources. If similarity beyond a reasonable amount is detected, penalties extending to filing an academic dishonesty case may be applied.
- 3- All students are expected to submit their work on time; late submissions without prior approved excuse will not be accepted.
- 4- Students that arrive late at class after 10 minutes will not be allowed entrance and will be marked absent.

WHILE EVERY EFFORT WILL BE MADE TO FOLLOW THE SCHEDULE, THE GRADING SCHEME AND THE ASSESSMENT DATE PLEASE NOTE THAT UNFORESEEN CIRCUMSTANCES CAN LEAD TO SOME CHANGES.